

Task 1:

A. Scatter Plots

A scatter plot depicts the relationship between two numerical variables by showing each observation as a point on the x and y axes. It determines if the association is positive, negative, or nonexistent, as well as how strong it is based on how closely the points cluster. Scatter plots are effective for identifying outliers and are frequently the initial step in correlation and regression research.

B. Bubble Charts

A bubble chart is an expansion of a scatter plot that incorporates a third variable, the size of each point, which is represented as bubbles. The x and y axes indicate two variables, while the bubble size (and color) adds additional dimensions to the data. These charts are handy for comparing numerous variables at once, but bubble sizes must be scaled appropriately.

C. Correlation Heatmaps

A correlation heatmap is a color-coded matrix that shows the correlations between numerous numerical variables. Each cell displays the correlation coefficient between two variables, with colors representing the strength and direction of the association. It is widely used in exploratory data analysis to quickly detect patterns, significant connections, and probable multicollinearity.

D. The Pearson Coefficient

The Pearson correlation coefficient (r) assesses the magnitude and direction of a linear relationship between two continuous variables. The value spans from -1 (perfect negative correlation) to +1 (perfect positive correlation), while 0 indicates no linear relationship. It is sensitive to outliers and can only detect linear patterns, not curved correlations.

E. Regression Analysis

Regression analysis uses a mathematical equation to model the relationship between a dependent variable and one or more independent variables. To minimize prediction errors, simple linear regression calculates the line of best fit ($y = mx + b$). This model can be used to produce predictions and is assessed using residuals to ensure accuracy and assumptions.

Task 2:

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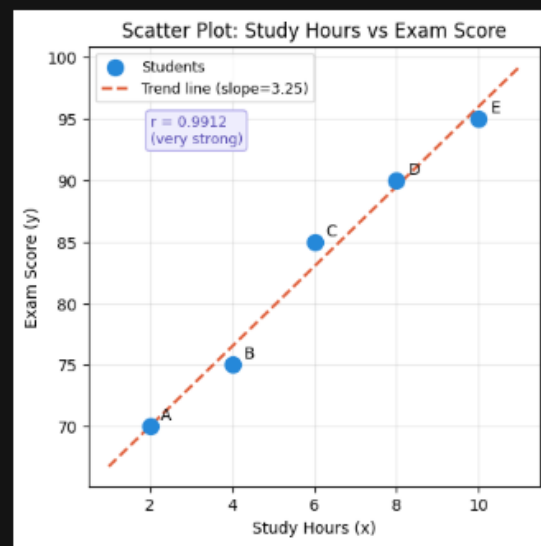
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TASK 2 - PEARSON CORRELATION COEFFICIENT
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student  x  y  x2  y2  xy
A  2  70  4  4900  140
B  4  75  16  5625  300
C  6  85  36  7225  510
D  8  90  64  8100  720
E  10 95  100 9025  950

n      = 5
Σx     = 30
Σy     = 415
Σx²    = 220
Σy²    = 34875
Σxy    = 2620

Numerator = 5×2620 - 30×415 = 650
Denominator = 655.7439
r        = 0.9912 (very strong correlation)
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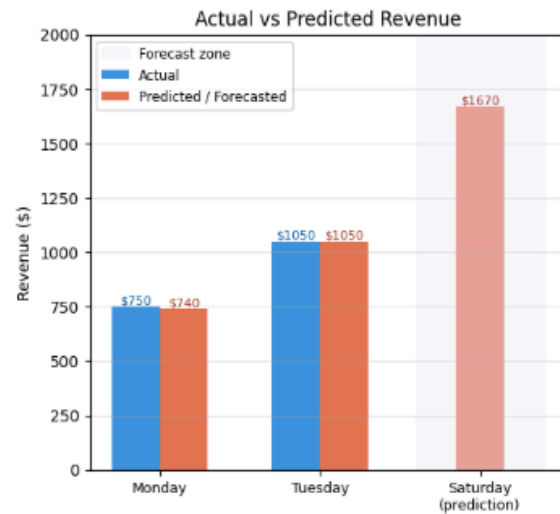
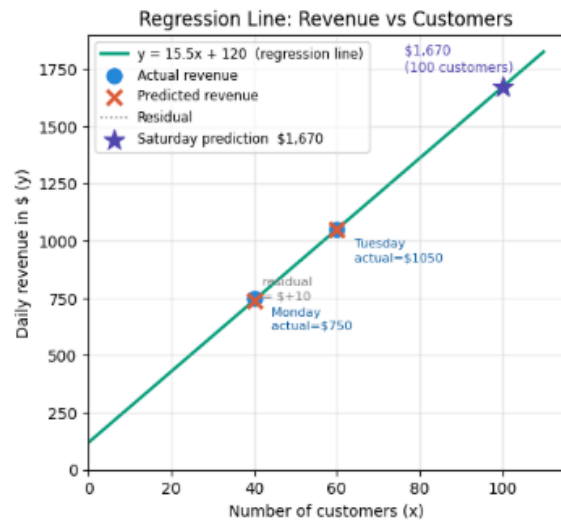
As seen on the line chart and the calculation using the Pearson r , there is a strong positive linear correlation between study hours because r is so near to +1 and test results. Practically speaking, students who regularly put in more study time typically get higher exam results. Every extra hour of study is linked to an approximately constant improvement in exam performance because the relationship is almost completely linear.

Task 3:

Day	Customers (x)	Predicted Revenue	Actual Revenue	Residual
Monday	40	\$740.0	\$750	\$+10.0
Tuesday	60	\$1,050.0	\$1,050	\$+0.0

The table shows that it has a positive residual which suggests that the model underestimated real revenue, whereas a negative residual means it overestimated. Both residuals are relatively modest, indicating that the model fits the data well.

Task 3 - Regression Analysis & Prediction



Given the equation of the Line of Best Fit: $y = 15.5x + 120$, if the coffee shop serves 100 customers on Saturday, the model predicts a daily revenue of \$1,670.0 which as seen on the bar graphs shows that the revenue increased linearly. Meanwhile, The slope of 15.5 reflects the average revenue generated for each extra customer. In real-world business terms, each new customer that comes into the coffee shop is expected to contribute. The total revenue for the day was roughly \$15.50. This statistic represents the average transaction value, which is the typical amount a consumer spends on coffee, food, or other things per visit. The intercept of 120, on the other hand, indicates the shop's baseline or fixed revenue independent of foot traffic, such as pre-arranged catering orders or standing contracts, even on a day with no walk-in customers.